

[Assignment VI]

Problem 2: Default weight initialization. Consider the multi-layer perceptron

$$\begin{aligned}
 y_L &= A_L y_{L-1} + b_L \\
 y_{L-1} &= \sigma(A_{L-1} y_{L-2} + b_{L-1}) \\
 &\vdots \\
 y_2 &= \sigma(A_2 y_1 + b_2) \\
 y_1 &= \sigma(A_1 x + b_1),
 \end{aligned}$$

$$y = \sum A_{L,j} y_{L-1,j} + b_L$$

where $x \in \mathbb{R}^{n_0}$, $A_\ell \in \mathbb{R}^{n_\ell \times n_{\ell-1}}$, $b_\ell \in \mathbb{R}^{n_\ell}$, and $n_L = 1$. For the sake of simplicity, let

$$\sigma(z) = z. \quad x \sim N(0, 1)$$

Assume $x_1, \dots, x_{n_0}$ are IID with zero-mean and unit variance. If this network is initialized with the default weight initialization of PyTorch, what will the mean and variance of y_L be?

PyTorch default initialization;

$$y_{\ell,i} = \sum_j A_{\ell,i,j} y_{\ell-1,j} + b_i, \quad A_{\ell,i,j} \sim \text{Uni}\left(-\sqrt{\frac{1}{n_{\ell-1}}}, \sqrt{\frac{1}{n_{\ell-1}}}\right), \quad b_i \sim \text{Uni}\left(-\frac{1}{\sqrt{n_\ell}}, \frac{1}{\sqrt{n_\ell}}\right)$$

$$\mathbb{E}(y_{\ell,i}) = \sum_j \mathbb{E}(A_{\ell,i,j} y_{\ell-1,j}) + \mathbb{E}(b_i)$$

$$= \sum_j \mathbb{E}(A_{\ell,i,j}) \mathbb{E}(y_{\ell-1,j}) + 0 = 0 \quad (\because \text{independence})$$

$$\Rightarrow \mathbb{E}(y_L) = \mathbb{E}(y_{L,1}) = 0$$

$$\text{Var}(y_{\ell,i}) = \text{Var}\left(\sum_j A_{\ell,i,j} y_{\ell-1,j} + b_i\right)$$

$$= \sum_j \text{Var}(A_{\ell,i,j} y_{\ell-1,j}) + \text{Var}(b_i) \quad (\because \text{independence})$$

$$= \sum_j \text{Var}(A_{\ell,i,j}) \text{Var}(y_{\ell-1,j}) + \text{Var}(b_i) \quad (\because \text{independence}) \quad (\text{Var}(y_{\ell-1,j}) \text{ 은 } j \text{ 에 따라 같지 않음})$$

$$= n_{\ell-1} \left(\frac{1}{n_{\ell-1}} \cdot \left(\sqrt{\frac{2}{n_{\ell-1}}} \right)^2 \right) \text{Var}(y_{\ell-1,1}) + \frac{1}{n_{\ell-1}} \left(\sqrt{\frac{2}{n_{\ell-1}}} \right)^2 = \frac{1}{3} \text{Var}(y_{\ell-1,1}) + \frac{1}{3n_{\ell-1}}$$

$$\circ \text{Var}(y_L) = \text{Var}(y_{L,1})$$

$$= \frac{1}{3} \text{Var}(y_{L-1,1}) + \frac{1}{3n_{L-1}}$$

$$= \frac{1}{3} \left(\frac{1}{3} \text{Var}(y_{L-2,1}) + \frac{1}{3n_{L-2}} \right) + \frac{1}{3n_{L-1}}$$

$$= \dots$$

$$= \left(\frac{1}{3} \right)^L \text{Var}(y_{0,1}) + \sum_{i=0}^{L-1} \left(\frac{1}{3} \right)^{L-i} \frac{1}{n_i}$$

$$= \left(\frac{1}{3} \right)^L + \sum_{i=0}^{L-1} \left(\frac{1}{3} \right)^{L-i} \frac{1}{n_i} //$$

→ 이항식 3/4항에 의해
평균은 생각할 필요 없음

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ମନେ ପକାନ୍ତି $\frac{dy}{dx}$ ମାନ କେବଳ x ଉପରେ ନିର୍ଭର କରେ,

$y^T A w_{\text{def}}$ 은 공변 노이즈, 이는 편지

(Weierstrass)의 정리는 다음과 같다.

Problem 4: Change of variables formula for Gaussians. If $\varphi: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a one-to-one differentiable function, $Y = \varphi(X)$, and Y is a continuous random variable with density function p_Y , then X is a continuous random variable with density function

$$p_X(x) = p_Y(\varphi(x)) \left| \det \frac{\partial \varphi}{\partial x}(x) \right|.$$

Let $Y \in \mathbb{R}^n$ be a continuous random vector with density

$$p_Y(y) = \frac{1}{(2\pi)^{n/2}} e^{-\frac{1}{2}\|y\|^2},$$

i.e., $Y \sim \mathcal{N}(0, I)$. Let $X = AY + b$ with an invertible matrix $A \in \mathbb{R}^{n \times n}$ and a vector $b \in \mathbb{R}^n$. Define $\Sigma = AA^T$. Show that X is a continuous random vector with density

$$p_X(x) = \frac{1}{\sqrt{(2\pi)^n \det \Sigma}} e^{-\frac{1}{2}(x-b)^T \Sigma^{-1}(x-b)}.$$

$$Y = A^{-1}(X-b) = \varphi(X) \Rightarrow \frac{\partial \varphi}{\partial X} = A^{-1}$$

$$\Sigma = AA^T \Rightarrow \det \Sigma = (\det A)^2, \quad |\det A| = \sqrt{\det \Sigma}, \quad |\det A^{-1}| = 1/\sqrt{\det \Sigma}$$

$$p_X(x) = p_Y(\varphi(x)) \left| \det \frac{\partial \varphi}{\partial x}(x) \right| = p_Y(\varphi(x)) |\det A^{-1}|$$

$$\rightarrow (A^{-1})^T A^{-1} = (AA^T)^{-1} = \Sigma^{-1}$$

$$= \frac{1}{(2\pi)^{n/2}} e^{-\frac{1}{2}\|A^{-1}(x-b)\|^2} \frac{1}{\sqrt{\det \Sigma}} = \frac{1}{(2\pi)^{n/2}} e^{-\frac{1}{2}(x-b)^T (A^{-1})^T A^{-1} (x-b)} \frac{1}{\sqrt{\det \Sigma}}$$

$$= \frac{1}{\sqrt{(2\pi)^n \det \Sigma}} e^{-\frac{1}{2}(x-b)^T \Sigma^{-1}(x-b)} //$$

Problem 5: D_{KL} of continuous random variables. The KL-divergence between continuous random variables $X \sim f$ and $Y \sim g$, where f and g are probability density functions in $\mathbb{R}^d$, is

$$D_{\text{KL}}(X \| Y) = \int_{\mathbb{R}^d} f(x) \log \left(\frac{f(x)}{g(x)} \right) dx.$$

(a) Show that

$$D_{\text{KL}}(X \| Y) \geq 0.$$

($\because -\log$, convex, Jensen)

$$D_{\text{KL}}(X \| Y) = \int_{\mathbb{R}^d} f(x) \log \frac{f(x)}{g(x)} dx = \int_{\mathbb{R}^d} f(x) \left(-\log \frac{g(x)}{f(x)} \right) dx \geq \int_{\mathbb{R}^d} -\log \left(f(x) \cdot \frac{g(x)}{f(x)} \right) dx = \int_{\mathbb{R}^d} -\log g(x) dx \geq 0 //$$

(b) Show that if $X = (X_1, \dots, X_d)$ is a continuous random variable such that $X_1, \dots, X_d$ are independent and $Y = (Y_1, \dots, Y_d)$ is a continuous random variable such that $Y_1, \dots, Y_d$ are independent, then

$$D_{\text{KL}}(X \| Y) = D_{\text{KL}}(X_1 \| Y_1) + \dots + D_{\text{KL}}(X_d \| Y_d).$$

$\log \frac{f(x)}{g(x)} = \dots + \log \frac{f(x_i)}{g(x_i)}$

$$D_{\text{KL}}(X \| Y) = \int_{\mathbb{R}^d} f(x) \log \frac{f(x)}{g(x)} dx = \int_{\mathbb{R}^d} f(x_1) \dots f(x_d) \left(\log \frac{f(x_1) \dots f(x_d)}{g(x_1) \dots g(x_d)} \right) dx_1 \dots dx_d$$

($\because$ independence)

$$= \sum_{i=1}^d \int_{\mathbb{R}^d} f(x_1) \dots f(x_d) \log \frac{f(x_i)}{g(x_i)} dx_1 \dots dx_d$$

$$= \sum_{i=1}^d \left(\int_{\mathbb{R}} f(x_i) \log \frac{f(x_i)}{g(x_i)} dx_i \right) \prod_{j \neq i} \left(\int_{\mathbb{R}} f(x_j) dx_j \right) = \sum_{i=1}^d \int_{\mathbb{R}} f(x_i) \log \frac{f(x_i)}{g(x_i)} dx_i = \sum_{i=1}^d D_{\text{KL}}(X_i \| Y_i) //$$

($\because$ independence)

Problem 6: D_{KL} of Gaussian random variables. Let $\mathcal{N}(\mu, \Sigma)$ denote the Gaussian distribution with mean μ and covariance Σ . So if $X \sim \mathcal{N}(\mu, \Sigma)$, then

$$\mathbb{E}[X] = \mu, \quad \mathbb{E}[(X - \mu)(X - \mu)^T] = \Sigma. \quad \Rightarrow \quad \mathbb{E}[\text{tr}(XX^T)]$$

$$= \text{tr}(\mathbb{E}(XX^T)) = \text{tr}(\Sigma + X\mu^T) = \text{tr}(\Sigma + \mu\mu^T)$$

Show that

$$D_{KL}(\mathcal{N}(\mu_0, \Sigma_0) \| \mathcal{N}(\mu_1, \Sigma_1)) = \frac{1}{2} \left(\text{tr}(\Sigma_1^{-1}\Sigma_0) + (\mu_1 - \mu_0)^T \Sigma_1^{-1}(\mu_1 - \mu_0) - d + \log \left(\frac{\det \Sigma_1}{\det \Sigma_0} \right) \right),$$

where d is the underlying dimension of the random variables $\mathcal{N}(\mu_0, \Sigma_0)$ and $\mathcal{N}(\mu_1, \Sigma_1)$.

$$D_{KL}(\mathcal{N}(\mu_0, \Sigma_0) \| \mathcal{N}(\mu_1, \Sigma_1)) = \mathbb{E}_{P_0}(\log P_0 - \log P_1)$$

$$= \frac{1}{2} \mathbb{E}_{P_0} \left[-(X - \mu_0)^T \Sigma_0^{-1} (X - \mu_0) + (X - \mu_1)^T \Sigma_1^{-1} (X - \mu_1) - \log \det \Sigma_0 + \log \det \Sigma_1 \right]$$

$$= \frac{1}{2} \log \frac{\det \Sigma_1}{\det \Sigma_0} + \frac{1}{2} \mathbb{E}_{P_0} \left[-(X - \mu_0)^T \Sigma_0^{-1} (X - \mu_0) + (X - \mu_1)^T \Sigma_1^{-1} (X - \mu_1) \right]$$

$$= \frac{1}{2} \log \frac{\det \Sigma_1}{\det \Sigma_0} + \frac{1}{2} \mathbb{E}_{P_0} \left[-\text{tr}(\Sigma_0^{-1} (X - \mu_0)(X - \mu_0)^T) + \text{tr}(\Sigma_1^{-1} (X - \mu_1)(X - \mu_1)^T) \right]$$

$$\because \langle U, W \rangle = U^T W = \text{tr}(UW^T)$$

$$= \frac{1}{2} \log \frac{\det \Sigma_1}{\det \Sigma_0} + \frac{1}{2} \left[-\text{tr}(\Sigma_0^{-1} \mathbb{E}_{P_0}((X - \mu_0)(X - \mu_0)^T)) + \text{tr}(\Sigma_1^{-1} \mathbb{E}_{P_0}((X - \mu_1)(X - \mu_1)^T)) \right]$$

$$= \frac{1}{2} \log \frac{\det \Sigma_1}{\det \Sigma_0} + \frac{1}{2} \left[-\text{tr}(\Sigma_0^{-1} \Sigma_0) + \text{tr}(\Sigma_1^{-1} (\Sigma_0 + \mu_0 \mu_0^T - 2\mu_0 \mu_1^T + \mu_1 \mu_1^T)) \right]$$

$$= \frac{1}{2} \left(\log \frac{\det \Sigma_1}{\det \Sigma_0} - d + \text{tr}(\Sigma_1^{-1} \Sigma_0) + \text{tr}(\Sigma_1^{-1} (\mu_0 \mu_0^T - 2\mu_0 \mu_1^T + \mu_1 \mu_1^T)) \right)$$

$$= \frac{1}{2} \left(\log \frac{\det \Sigma_1}{\det \Sigma_0} - d + \text{tr}(\Sigma_1^{-1} \Sigma_0) + \text{tr}(\Sigma_1^{-1} (\mu_0 - \mu_1)(\mu_0 - \mu_1)^T) \right)$$

$$= \frac{1}{2} \left(\log \frac{\det \Sigma_1}{\det \Sigma_0} - d + \text{tr}(\Sigma_1^{-1} \Sigma_0) + (\mu_0 - \mu_1)^T \Sigma_1^{-1} (\mu_0 - \mu_1) \right)$$